KrishiMitra AI

AI, Generative AI, and RAG — Study Notes

1. Artificial Intelligence (AI)

Artificial Intelligence is a field of computer science focused on creating systems that perform tasks such as understanding language, recognizing patterns, and answering questions.

2. Generative AI

Generative AI uses models to create content such as text, images, audio, or code in response to an input or prompt.

3. Large Language Models (LLMs)

An LLM is an AI model trained to process and generate human language. It can explain concepts, summarize information, and answer questions, but it can sometimes produce incorrect answers.

4. Retrieval-Augmented Generation (RAG)

RAG combines information retrieval with language generation. It retrieves relevant information from external sources and provides it to an LLM as context for answering a question.

5. How KrishiMitra AI Can Use RAG

KrishiMitra AI is an AI-powered agricultural assistant concept. It can retrieve relevant information from trusted agricultural guides, farming documents, research papers, and government resources, then use an LLM to generate context-aware answers to farmers' questions.

6. Possible Knowledge Sources

PDF, TXT, Markdown, DOCX, agricultural websites, research papers, CSV and Excel files, scanned documents processed with OCR, available YouTube transcripts, manually entered text, and authorized agricultural APIs or databases.

7. Basic RAG Pipeline

1. Load documents or collect information.
2. Extract readable text and metadata.
3. Split text into chunks.
4. Create embeddings for chunks.
5. Store vectors and text in a vector store.
6. Retrieve chunks relevant to a question.
7. Add retrieved context and the question to a prompt.
8. Ask the LLM to generate an answer based on the supplied context.
9. Show source references when available.

8. Important Limitations

RAG does not normally train the LLM on uploaded documents. Retrieval and generation can still be incorrect, so reliable and up-to-date sources matter. Weather and market prices change frequently; live APIs may be better than a static document index. Agricultural guidance should be checked against trusted local expert or official advice when decisions affect crops, livestock, or finances.